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Academia Xideral

Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which of the following are legal entry point methods that can be run from the command line? (Choose all that apply.)

A. `private static void main(String[] args)`

B. `public static final main(String[] args)`

C. `public void main(String[] args)`

D. `public static final void main(String[] args)` Un método de tipo main sí puede ser final.

E. `public static void main(String[] args)` Este es el método predeterminado en java

F. `public static main(String[] args)`

2. Which answer options represent the order in which the following statements can be assembled into a program that will compile successfully? (Choose all that apply.)

```
X: class Rabbit {}
Y: import java.util.*;
Z: package animals;
```

A. X, Y, Z

B. Y, Z, X

C. Z, Y, X El órden debe ser package, import y clase.

D. Y, X El órden es correcto, primero import y posteriormente la clase

E. Z, X El órden es correcto, primero package y posteriormente la clase.

F. X, Z

G. None of the above

3. Which of the following are true? (Choose all that apply.)

```
public class Bunny {
   public static void main(String[] x) {
      Bunny bun = new Bunny();
} }
```

A. Bunny is a class. Bunny sí es una clase. "public class Bunny"

B. bun is a class.

C. `main` is a class.

D. Bunny is a reference to an object.

E. bun is a reference to an object. Bun es una variable de referencia que apunta a un objeto.

F. `main` is a reference to an object.

G. The `main()` method doesn't run because the parameter name is incorrect.

4. Which of the following are valid Java identifiers? (Choose all that apply.)

A. `_`

B. `_helloWorld$` Sí se puede utilizar "_" y "$" siempre y cuando estén acompañados de otros caracteres válidos

C. `true`

D. `java.lang`

E. `Public` "Public" no es una palabra reservada en java, pero "public" sí.

F. `1980_s`

G. `_Q2_` Es totalmente válido utilizar "_" acompañado al principio o al final de otros caracteres.

5. Which statements about the following program are correct? (Choose all that apply.)

```
2:  public class Bear {
3:     private Bear pandaBear;
4:     private void roar(Bear b) {
5:        System.out.println("Roar!");
6:        pandaBear = b;
7:     }
8:     public static void main(String[] args) {
9:        Bear brownBear = new Bear();
10:       Bear polarBear = new Bear();
11:       brownBear.roar(polarBear);
12:       polarBear = null;
13:       brownBear = null;
14:       System.gc(); } }
```

A. The object created on line 9 is eligible for garbage collection after line 13. En la línea 13 cuando brownBear es null, ya no apunta a ningún objeto.

B. The object created on line 9 is eligible for garbage collection after line 14.

C. The object created on line 10 is eligible for garbage collection after line 12.

D. The object created on line 10 is eligible for garbage collection after line 13. A pesar de que polarBear ya no apunta a nada en la línea 12. Aún brownBear.pandaBear puede acceder al objeto de la línea 10. Cuando se elimina brownBear, se eliminan todas sus referencias.

E. Garbage collection is guaranteed to run.

F. Garbage collection might or might not run. Cierto, no siempre que se llama puede correr el garbage collector.

G. The code does not compile.

6. Assuming the following class compiles, how many variables defined in the class or method are in scope on the line marked on line 14?

```
1:  public class Camel {
2:     { int hairs = 3_000_0; }
3:     long water, air=2;
4:     boolean twoHumps = true;
5:     public void spit(float distance) {
6:        var path = "";
```

Las variables activades para el campo estan marcadas de azul.

```
7:         { double teeth = 32 + distance++; }
8:         while(water > 0) {
9:            int age = twoHumps ? 1 : 2;
10:           short i=-1;
11:           for(i=0; i<10; i++) {
12:              var Private = 2;
13:           }
14:           // SCOPE
15:        }
16:     }
17: }
```

A. 2

B. 3

C. 4

D. 5

E. 6

F. 7

G. None of the above

7. Which are true about this code? (Choose all that apply.)

```
public class KitchenSink {
    private int numForks;

    public static void main(String[] args) {
       int numKnives;
       System.out.print("""
          "# forks = " + numForks +
           " # knives = " + numKnives +
          # cups = 0""");
    }
}
```

A. The output includes: `# forks = 0`.

B. The output includes: `# knives = 0`.

C. The output includes: `# cups = 0`. Como es una cadena, imprimirá exactamente "# cups =0"

D. The output includes a blank line.

E. The output includes one or more lines that begin with whitespace. Justo cuando se hace un tab en "# cups" abrá un espacio en blanco al principio.

F. The code does not compile.

8. Which of the following code snippets about `var` compile without issue when used in a method? (Choose all that apply.)

A. `var spring = null;`

B. `var fall = "leaves";` Forma idónea de inicializar una variable var, indicándole el tipo que es con una asignación.

C. `var evening = 2; evening = null;`

D. `var night = Integer.valueOf(3);` Aquí la función de Integer le permite inferir que lo que espera es un integer.

E. `var day = 1/0;` A pesar de que esto dará un error en runtinme, el ejercicio especificó que fuera en tiempo de compilación.

F. `var winter = 12, cold;`

G. `var fall = 2, autumn = 2;`

H. `var morning = ""; morning = null;` Según el libro, sí puede inicializarse primero una variable NO PRIMITIVA y Posteriormente darle null, puesto que el compilador ya determinó su tipo.

9. Which of the following are correct? (Choose all that apply.)

A. An instance variable of type `float` defaults to `0`.

B. An instance variable of type `char` defaults to `null`.

C. A local variable of type `double` defaults to `0.0`.

D. A local variable of type `int` defaults to `null`.

E. A class variable of type `String` defaults to `null`. Si no es un primitivo, su valor por defecto siempre será null.

F. A class variable of type `String` defaults to the empty string `""`. Java nunca crea objetos automáticamente para rellenar variables! Esto es un error!

G. None of the above.

10. Which of the following expressions, when inserted independently into the blank line, allow the code to compile? (Choose all that apply.)

```
public void printMagicData() {
   var magic = ______________;
   System.out.println(magic);
}
```

A. `3_1`

B. `1_329_.0`

C. `3_13.0_`

D. `5_291._2`

E. `2_234.0_0`

F. `9___6`

G. `_1_3_5_0`

Los "_" no pueden ponerse antes o después de un punto. Así como no se puede colocar al principio o al final.

11. Given the following two class files, what is the maximum number of imports that can be removed and have the code still compile?

```
// Water.java
package aquarium;
public class Water { }
```

```
// Tank.java
package aquarium;
import java.lang.*;
import java.lang.System;
import aquarium.Water;
import aquarium.*;
public class Tank {
   public void print(Water water) {
   System.out.println(water); } }
```

A. 0

B. 1

C. 2

D. 3

E. 4

F. Does not compile

Tanto "import java.lang*;" como "import java.lang.System" son bibliotecas que java importa por defecto. "aquarium.Water;" y "aquarium.*" Ya se importan directamente con el Package de "aquarium".

12. Which statements about the following class are correct? (Choose all that apply.)

```
1: public class ClownFish {
2:    int gills = 0, double weight=2;
3:    { int fins = gills; }
4:    void print(int length = 3) {
5:       System.out.println(gills);
6:       System.out.println(weight);
7:       System.out.println(fins);
8:       System.out.println(length);
9: } }
```

A. Line 2 generates a compiler error.

No se separa entre comas, sino entre ";"

B. Line 3 generates a compiler error.

C. Line 4 generates a compiler error.

Java no deja inicializar variables en los parámetros como en Python

D. Line 7 generates a compiler error.

El alcance de fins no es correcto.

E. The code prints `0`.

F. The code prints `2.0`.

G. The code prints `2`.

H. The code prints `3`.

13. Given the following classes, which of the following snippets can independently be inserted in place of `INSERT IMPORTS HERE` and have the code compile? (Choose all that apply.)

```
package aquarium;
public class Water {
   boolean salty = false;
}

package aquarium.jellies;
public class Water {
   boolean salty = true;
}

package employee;
INSERT IMPORTS HERE
public class WaterFiller {
   Water water;
}
```

A. `import aquarium.*;`

Podemos importar todo, es correcto.

B. `import aquarium.Water;`
`import aquarium.jellies.*;`

Aquarium ya importa water y previamente jellies ya es importado también.

C. `import aquarium.*;`
`import aquarium.jellies.Water;`

Esta es correcta, porque importamos todo antes y también podemos importar jellies.water

D. `import aquarium.*;`
`import aquarium.jellies.*;`

E. `import aquarium.Water;`
`import aquarium.jellies.Water;`

F. None of these imports can make the code compile.

14. Which of the following statements about the code snippet are true? (Choose all that apply.)

```
3: short numPets = 5L;
4: int numGrains = 2.0;
5: String name = "Scruffy";
6: int d = numPets.length();
7: int e = numGrains.length;
8: int f = name.length();
```

Forma correcta de declarar una string.

fallé, es un tipo primitivo y no debe llamar a una función.

Dado que name arriba es de tipo String, entonces sí podemos usar la función length()

A. Line 3 generates a compiler error.

B. Line 4 generates a compiler error.

C. Line 5 generates a compiler error.

D. Line 6 generates a compiler error.

E. Line 7 generates a compiler error.

F. Line 8 generates a compiler error.

15. Which of the following statements about garbage collection are correct? (Choose all that apply.)

A. Calling `System.gc()` is guaranteed to free up memory by destroying objects eligible for garbage collection.

B. Garbage collection runs on a set schedule.

C. Garbage collection allows the JVM to reclaim memory for other objects. Efectivamente java utiliza el Garbage Collection para liberar espacio en memoria.

D. Garbage collection runs when your program has used up half the available memory.

E. An object may be eligible for garbage collection but never removed from the heap. Sí, un objeto podrías ser candidato para ser borrado.

F. An object is eligible for garbage collection once no references to it are accessible in the program. Sí, cuando ya no hay referencias que apunten al objeto es elegible para ser borrado.

G. Marking a variable `final` means its associated object will never be garbage collected. Fallé :)

16. Which are true about this code? (Choose all that apply.)

```
var blocky = """
   squirrel \s
   pigeon   \
   termite""";
System.out.print(blocky);
```

A. It outputs two lines. El "\" significa que será sobre la misma línea sin tap.

B. It outputs three lines.

C. It outputs four lines.

D. There is one line with trailing whitespace. \s le dice a java que guarde los espacios de la cola.

E. There are two lines with trailing whitespace.

F. If we indented each line five characters, it would change the output.

17. What lines are printed by the following program? (Choose all that apply.)

```
1:  public class WaterBottle {
2:     private String brand;
3:     private boolean empty;
4:     public static float code;
5:     public static void main(String[] args) {
6:        WaterBottle wb = new WaterBottle();
```

```
7:        System.out.println("Empty = " + wb.empty);
8:        System.out.println("Brand = " + wb.brand);
9:        System.out.println("Code = " + code);
10:    } }
```

A. Line 8 generates a compiler error.

B. Line 9 generates a compiler error.

C. `Empty =`

D. `Empty = false`

E. `Brand =`

F. `Brand = null`

G. `Code = 0.0`

H. `Code = 0f`

Aquellas variables que son primitivas se inicializan con un valor predeterminado.

18. Which of the following statements about `var` are true? (Choose all that apply.)

A. A `var` can be used as a constructor parameter.

B. The type of a `var` is known at compile time. Sí, por eso es importante inicializar a la variable var.

C. A `var` cannot be used as an instance variable. No puedes usarlo para definir atributos de una clase.

D. A `var` can be used in a multiple variable assignment statement.

E. The value of a `var` cannot change at runtime.

F. The type of a `var` cannot change at runtime. La clase de "var" se infiere en runtime, así que no se puede cambiar

G. The word `var` is a reserved word in Java.

19. Which are true about the following code? (Choose all that apply.)

```
var num1 = Long.parseLong("100");
var num2 = Long.valueOf("100");
System.out.println(Long.max(num1, num2));
```

A. The output is `100`. Ya sea por num1 o num2 el max será 100.

B. The output is `200`.

C. The code does not compile.

D. `num1` is a primitive. Cuando parseamos a long num1, var se vuelve de tipo long primitiva.

E. `num2` is a primitive.

20. Which statements about the following class are correct? (Choose all that apply.)

```
1:  public class PoliceBox {
2:     String color;
3:     long age;
4:     public void PoliceBox() {
5:        color = "blue";
6:        age = 1200;
```

```
7:     }
8:     public static void main(String []time) {
9:        var p = new PoliceBox();
10:       var q = new PoliceBox();
11:       p.color = "green";
12:       p.age = 1400;
13:       p = q;
14:       System.out.println("Q1="+q.color);
15:       System.out.println("Q2="+q.age);
16:       System.out.println("P1="+p.color);
17:       System.out.println("P2="+p.age);
18: } }
```

A. It prints `Q1=blue`.

B. It prints `Q2=1200`.

C. It prints `P1=null`. Puesto que nunca se inicializó P1.

D. It prints `P2=1400`.

E. Line 4 does not compile.

F. Line 12 does not compile.

G. Line 13 does not compile.

H. None of the above.

21. What is the output of executing the following class?

```
1:  public class Salmon {
2:     int count;
3:     { System.out.print(count+"-"); }
4:     { count++; }
5:     public Salmon() {
6:        count = 4;
7:        System.out.print(2+"-");
8:     }
9:     public static void main(String[] args) {
10:       System.out.print(7+"-");
11:       var s = new Salmon();
12:       System.out.print(s.count+"-"); } }
```

A. `7-0-2-1-`

B. `7-0-1-`

C. `0-7-2-1-`

D. `7-0-2-4-`

E. `0-7-1-`

1.- Se imprime el "7-" del main.
2.- Se imprime count = 0 y "-"
3.- se aumenta en 1 count
4.- count vale 4
5.- se imprime "2-"
6.- se imprime count con valor 4 y -.

F. The class does not compile because of line 3.

G. The class does not compile because of line 4.

H. None of the above.

22. Given the following class, which of the following lines of code can independently replace `INSERT CODE HERE` to make the code compile? (Choose all that apply.)

```
public class Price {
   public void admission() {
      INSERT CODE HERE
      System.out.print(amount);
      } }
```

A. `int Amount = 0b11;`

B. `int amount = 9L;`

C. `int amount = 0xE;` Representación de valor hexadecimal

D. `int amount = 1_2.0;`

E. `double amount = 1_0_.0;`

F. `int amount = 0b101;` Representación del valor binario 101

G. `double amount = 9_2.1_2;` Representación correcta del double 92.12

H. `double amount = 1_2_.0_0;`

23. Which statements about the following class are true? (Choose all that apply.)

```
1:  public class River {
2:     int Depth = 1;
3:     float temp = 50.0;
4:     public void flow() {
5:        for (int i = 0; i < 1; i++) {
6:           int depth = 2;
7:           depth++;
8:           temp--;
9:        }
```

```
10:       System.out.println(depth);
11:       System.out.println(temp); }
12:    public static void main(String... s) {
13:       new River().flow();
14: } }
```

A. Line 3 generates a compiler error. Le falta la "f" al final del float.

B. Line 6 generates a compiler error.

C. Line 7 generates a compiler error.

D. Line 10 generates a compiler error. depth está fuera de ámbito

E. The program prints `3` on line 10.

F. The program prints `4` on line 10.

G. The program prints `50.0` on line 11.

H. The program prints `49.0` on line 11.